

JINYU ZHANG

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Date of Birth: Apr 2001

EDUCATION

École Polytechnique Fédérale de Lausanne (EPFL) Lausanne, Switzerland
Ph.D. Student in Civil and Environmental Engineering Sep 2026–present
Urban Transport Systems Laboratory (LUTS)
Advisor: Prof. Nikolas Geroliminis

Southeast University Nanjing, China
M.Eng. in Traffic and Transportation Engineering Sep 2023–Jun 2026
Advisor: Prof. Di Huang, Prof. Zhiyuan Liu

Beijing Jiaotong University Beijing, China
B.Eng. in Traffic Engineering Sep 2019–Jun 2023

RESEARCH TOPICS

Contextual Optimization; Learning to Optimize

SELECTED PUBLICATIONS

Journal Publications

1. **Zhang, J.**, Huang, D.*, Liu, Z., et al. (2024). A data-driven optimization-based approach for freeway traffic state estimation based on heterogeneous sensor data fusion. *Transportation Research Part E*, 189, 103656. <https://doi.org/10.1016/j.tre.2024.103656>.
2. Huang, D., **Zhang, J.**, & Liu, Z.* (2023). A robust coordinated charging scheduling approach for hybrid electric bus charging systems. *Transportation Research Part D*, 125, 103955. <https://doi.org/10.1016/j.trd.2023.103955>.
3. Huang, D., **Zhang, J.**, Liu, Z.*, et al. (2024). A novel ranking method based on semi-SPO for battery swapping allocation optimization in a hybrid electric transit system. *Transportation Research Part E*, 188, 103611. <https://doi.org/10.1016/j.tre.2024.103611>.
4. Huang, D., **Zhang, J.**, Liu, Z.*, & Liu, R. (2025). Prescriptive analytics for freeway traffic state estimation by multi-source data fusion. *Transportation Research Part E*, 198, 104105. <https://doi.org/10.1016/j.tre.2025.104105>.

Conference Presentations

1. **Zhang, J.**, Huang, D.*, Liu, Z., & Zheng, Y. (2024). A fine-grained full-coverage highway traffic flow condition perception method based on heterogeneous sensor data fusion. *The 2024 Transportation Research Board (TRB) Annual Meeting*.

- al Meeting*, Washington D.C., **Poster**. <https://annualmeeting.mytrb.org/OnlineProgramArchive/Details/21093>.
2. **Zhang, J.**, Huang, D.*, & Liu, Z. (2025). Prescriptive analytics for freeway traffic state estimation based on data fusion. *The 2025 Transportation Research Board (TRB) Annual Meeting*, Washington D.C., **Poster**. <https://annualmeeting.mytrb.org/OnlineProgramArchive/Details/22757>.
 3. **Zhang, J.**, Huang, D.*, & Liu, Z. (2025). A data-driven optimization-based approach for freeway traffic state estimation based on heterogeneous sensor data fusion. *The 2024 World Transport Convention (WTC)*, Qingdao, **Oral**.
 4. **Zhang, J.**, Huang, D.*, & Liu, Z. Prescriptive analytics for freeway traffic state estimation based on data fusion. *The 6th International Symposium on Multimodal Transportation*, Nanjing, **Oral**.

SELECTED ACADEMIC PROJECTS

1. **Incremental Efficient Production-Scheduling for Supply Chains**
 - Enhanced the relax-and-fix algorithm for ultra-large-scale production scheduling; derived strategies with theoretical guarantees and achieved 10% end-to-end acceleration.
 - Developed an AI-for-OR approach tailored to problem structure, delivering 30% end-to-end acceleration.
2. **Deterministic Parallel Branch-and-Bound Tree in MIP Problems**
 - Designed a multi-threaded deterministic parallel algorithm for mixed-integer programming.
 - Implemented the method in the open-source solver HiGHS; achieved 1.9× speedup with 8 threads on Benchmark2017.

SELECTED HONORS & AWARDS

- **2026** Outstanding Graduate, Southeast University
- **2025** FTTE Future Stars Award
- **2025** Southeast University Presidential Scholarship
- **2025** Xiaomi Grand Scholarship
- **2025** Pacesetter Model Graduate Student of Southeast University
- **2025** National Scholarship of China
- **2024** National Scholarship of China
- **2024** Academic Innovation Excellence Award
- **2024** First-Class Academic Scholarship
- **2024** Southeast University Traffic Design Institute Scholarship

TEACHING EXPERIENCE

1. School of Transportation, Southeast University Mar 2024–Jun 2024
Advanced Operations Research, *Teaching Assistant*.
2. School of Transportation, Southeast University Mar 2025–Jun 2025
Transport Systems Analysis, *Teaching Assistant*.